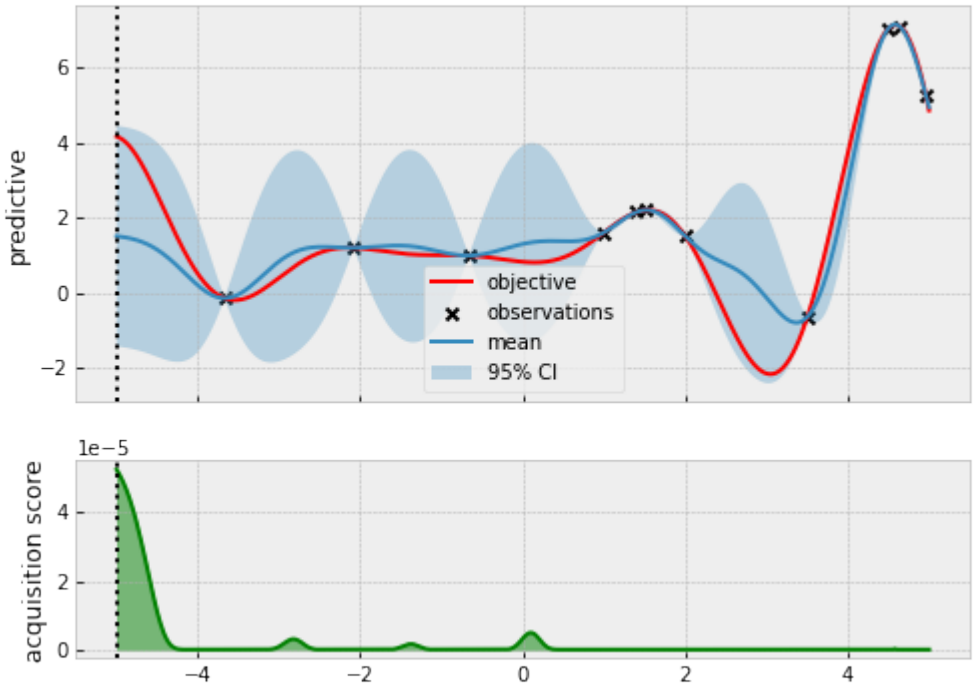


Figure A.5 Optimization progress of the modified Probability of Improvement at the last iteration. The policy has found the optimum.

Here, the modified PoI has found the optimum.

5. How much more explorative PoI becomes heavily depends on the minimum improvement threshold stored in `epsilon`. Setting this variable to 0.001 doesn't sufficient encourage exploration, and the policy once again gets stuck. Setting this variable to 0.5 works well.
6. Implement a relative minimum improvement threshold with a 110% improvement requirement.

```
epsilon_pct = 0.1

for i in range(num_queries):
    ... ❶

    policy = botorch.acquisition.analytic.ProbabilityOfImprovement(
        model, best_f=train_y.max() * (1 + epsilon_pct) ❷
    )
```

❶ Omitted.

❷ Relative improvement.

A.4.2 Bayesian optimization for hyperparameter tuning

This exercise, implemented in `CH04/03 - Exercise 2.ipynb`, applies BayesOpt to an objective function that simulates the accuracy surface of a support-vector machine model in a hyperparameter tuning task. There are multiple steps:

1. Recreate the Bayesian optimization (BayesOpt) loop in `CH04/01 - BayesOpt loop`, which uses the one-dimensional Forrester function as the optimization objective. Our

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